

ELITE IGCSE MATHEMATICS  
EXPERIENCE NOTES

# Algebraic Proof

*Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.*

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ALGEBRAIC PROOF · CH2 · L8



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# The Map

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This lesson collapses Algebraic Proof into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

## 01 Start from the expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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## 02 Expand and collect

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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## 03 Use parity or multiple language

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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## 04 Finish with the required statement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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## 05 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

# 01 Start from the expression

**TRIGGER** *"work out", "show", "hence", a familiar exam phrase*

**BECOMES** A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

**FIRST LINE TO WRITE**

$$x = 0.5727272 \dots$$

**SIMPLEST STRATEGY**

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

**WORKED MODEL**

**Source pattern:** Jan 2020 P1HR Q13

**(a) Let**

$$x = 0.5727272 \dots$$

**Calculate value**

$$1000x = 572.727272 \dots$$

**Finish:** (a) shown, (b)  $\frac{6+3\sqrt{y}}{4-y}$

**FORMULA BANK**

Start with an expression, expand/simplify until the required form appears.

**BOTTOM LINE**

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

# Practice 01

**SAME IDEA** Use the same first line from Strategy 01.

## QUESTION

13 (a) Use algebra to show that  $0.57 \cdot 2 = 63 \cdot 110$  (2) Given that  $y$  is a prime number, (b) express  $3 \cdot 2 - y$  in the form  $a \cdot b \cdot y \cdot c \cdot y + \dots$  where  $a, b$  and  $c$  are integers. (2)

## YOUR SOLUTION

## 02 Expand and collect

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2020 P2H Q17

**Let the two consecutive square**

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

**is one more than an**

This is a reasoning step: write one clear sentence, then continue the calculation.

**Finish:** The difference is always odd.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

**SAME IDEA** Use the same first line from Strategy 02.

## QUESTION

17 Prove that the difference between two consecutive square numbers is always an odd number. Show clear algebraic working.

## YOUR SOLUTION

## 03 Use parity or multiple language

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$N = 5k$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2020 P2HR Q25

**Use trigonometry**

$$N = 5k$$

**where is an integer**

This is a reasoning step: write one clear sentence, then continue the calculation.

**Finish:**  $A^2 - B^2$  is always a multiple of 20.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.





## 04 Finish with the required statement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$N = 5k$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2020 P2HR Q25

**Use trigonometry**

$$N = 5k$$

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This is a reasoning step: write one clear sentence, then continue the calculation.

**Finish:**  $A^2 - B^2$  is always a multiple of 20.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

**SAME IDEA** Use the same first line from Strategy 04.

## QUESTION

25 N is a multiple of 5  $A = N + 1$   $B = N - 1$  Prove, using algebra, that  $A^2 - B^2$  is always a multiple of 20

## YOUR SOLUTION

## 05 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x = 0.5727272 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2020 P1HR Q13

**(a) Let**

$$x = 0.5727272 \dots$$

**Calculate value**

$$1000x = 572.727272 \dots$$

**Finish:** (a) shown, (b)  $\frac{6+3\sqrt{y}}{4-y}$

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

**SAME IDEA** Use the same first line from Strategy 05.

## QUESTION

13 (a) Use algebra to show that  $0.57 \cdot 2 = 63 \cdot 110$  (2) Given that  $y$  is a prime number, (b) express  $3 \cdot 2 - y$  in the form  $a \cdot b \cdot y \cdot c \cdot y + -$  where  $a, b$  and  $c$  are integers. (2)

## YOUR SOLUTION

# Quick Reference

TRIGGER → FIRST MOVE

| WHEN YOU SEE                                              | WRITE THIS FIRST                   |
|-----------------------------------------------------------|------------------------------------|
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Start from the expression          |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Expand and collect                 |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Use parity or multiple language    |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Finish with the required statement |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Subtract                           |

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.